Lab 14 — Build a Moving-Average Technical Analysis

Python Programming for Finance · TGS-2026065049 · LO6 / A2–A3 / A6–A9

Scenario

A quantitative analyst needs an OHLC view and 10/50-day moving averages with objectively detected crossover dates.

Goal

An offline-safe technical-analysis report with OHLC, moving averages and crossover evidence.

Tools

Pandas, Matplotlib, optional yfinance

Step-by-step

1. Load the supplied DBS OHLC data; optionally refresh with yfinance.

2. Validate required OHLC columns and date ordering.

3. Calculate MA10 and MA50 with rolling().mean().

```
python starter.py
```

4. Detect sign changes in MA10 minus MA50.

5. Plot Close, MA10 and MA50 with crossover markers.

```
python solution.py
```

6. Save crossover_dates.csv and explain the first crossover.

Acceptance criteria

The output contains OHLC rows, MA10/MA50 values, at least one crossover date, and a labelled chart; the offline file always works.

Evidence

Capture the executed output, validation result, generated file or chart, and a one-sentence finance interpretation.